

Yuanhao Zheng

Email: yuenhozheng@gmail.com | Tel: +86-188-8892-0546 | Web: zheng-yuanhao.github.io

Research Interests: Food quality, AI-enabled food system

EDUCATION

Zhejiang University (ZJU) *Hangzhou, China*

M.Eng. in Agricultural Engineering | [Grade: 88.6/100](#) *09/2022 – 06/2025*

Agricultural Systems Modeling and Big Data Analytics (93), Bio-systems Detection and Control Technologies (93), Engineering Analysis of Biological System (98), Special Topic of Precision Farming Technology (88)

Zhejiang University (ZJU) *Hangzhou, China*

B.Eng. in Food Science & Engineering | [GPA: 3.61/4.0](#) *09/2017 – 06/2021*

Analytical Chemistry (89), Organic Chemistry (80), Postharvest Technology of Fresh Fruit & Vegetables (89), Food Microbiology Analysis (89), Fermentation Technology (95), Food Additives (92), Fundamental of Data Structures (94)

PUBLICATIONS

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- [1] **Zheng, Y.**, Luo, X., Gao, Y., Sun, Z., Huang, K., Gao, W., Xu, H., Xie, L.* Lycopene detection in cherry tomatoes with feature enhancement and data fusion. *Food Chem.*, 2025. [[DOI](#)]
- [2] **Zheng, Y.**, Liu, P., Zheng, Y., et al. Improving SSC detection accuracy of cherry tomatoes by feature synergy and complementary spectral bands combination. *Postharvest Biol. Technol.*, 2024. [[DOI](#)]
- [3] **Zheng, Y.**, Zhou, Y., Liu, P., et al. Improving discrimination accuracy of pest-infested crabapples using Vis/NIR spectral morphological features. *J. Food Meas. Charact.*, 2024. [[DOI](#)]
- [4] Li, J., **Zheng, Y.**, Wang, P.*, Zhang, H.* The alginate dialdehyde crosslinking on curcumin-loaded zein nanofibers for controllable release. *Food Res. Int.*, 2024. [[DOI](#)]
- [5] Liu, P., **Zheng, Y.**, et al. Enhancing fruit SSC detection accuracy via a light attenuation theory-based correction method to mitigate measurement orientation variability. *Food Res. Int.*, 2024. [[DOI](#)]
- [6] Patent: Xie L., Liu P., **Zheng Y.**, Ying Y. Method for Improving Detection Accuracy by Matching Light Source Spectrum with Component Absorption Features. ZL 202410297919.5, 2025.

RESEARCH EXPERIENCE

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- [1] **Nondestructive detection of lycopene content in cherry tomatoes** *06/2022 – 06/2025*
- Improved lycopene detection accuracy in cherry tomatoes through UV/Vis/NIR spectral enhancement and spectra-image fusion ($R^2 = 0.95$).
 - Enhanced characteristic spectral bands and fused multi-source data for explainable interpretation.
- [2] **Soluble solids content (SSC) prediction of cherry tomatoes** *03/2023 - 09/2024*
- Improved SSC prediction performance using UV/Vis/NIR spectroscopy ($R^2 = 0.97$).
 - Identified sugar-related absorption spectral bands and aligned them with light source spectra.
 - Proposed the feature synergy and spectral bands combination strategy for SSC prediction.

[3] Pest-infested crabapples discrimination using Vis/NIR spectroscopy 10/2023 - 08/2024

- Compensated for the effects of fruit orientation by developing a global model integrating multi-position spectra, improving classification robustness.
- Discriminated pest-infested crabapples by enhancing spectral morphological feature between healthy and pest-infested crabapples' spectra (ACC = 95.94%).

[4] The crosslinked electrospun nanofibers for controllable curcumin release

(Undergraduate Thesis)

10/2020 – 06/2021

- Fabricated zein/alginate dialdehyde (AD) electrospun nanofibers via green crosslinking.
- Enhanced mechanical properties and controlled curcumin release for curcumin (CUR) encapsulation with improved antioxidant behavior.

ACADEMIC ACTIVITIES

Services

- Part-time counselor for undergraduate students in 1st year 08/2023 – 07/2024
- Peer review service for *Food Control* 05/2024 – present

Awards

- Excellent Postgraduate Students' Award 2024 – 2025
- National Scholarship (top 2%) 2023 – 2024
- Five-Excellence Graduate Student of Zhejiang University 2023 – 2024
- Award of Honor for Graduate, ZJU 2023 – 2024
- National Encouragement Scholarship 2018 – 2019

Conference Presentations

- Jul. 2024: Lycopene detection in cherry tomatoes with feature enhancement and data fusion, 13th Applied Optics and Photonics China (**AOPC 2024**), Oral, Beijing, China.
- May. 2024: Improving SSC detection accuracy of cherry tomatoes by feature synergy and spectral combination, the 6th CIGR International Conference 2024 (**CIGR 2024**), Oral, Jeju, South Korea.
- Aug. 2023: Nondestructive detection of soluble solids and lycopene content in cherry tomatoes using NIRS, Chinese Society of Agricultural Engineering (**CSAE 2023**), Poster, Chengdu, China.

SKILLS

Language: Mandarin (native), English (TOEFL98)

Program Skills: Python, MATLAB, C/C++

Experimental Skills: Machine/Deep Learning, Chemometrics, High-dimensional Data Analysis, Spectroscopy, Experimental Design, Statistical Analysis, Food Quality Evaluation, Chemical Analysis